Managya Singhal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.23
MILESTONE ACADEMY SENIOR SECONDARY SCHOOL JAORA	Class XII (MPBSE)	2024	83.20%
ST PETER'S HIGH SCHOOL JAORA	Class X (CBSE)	2022	94.60%

SKILLS

Programming Languages: C++, Python, C, SQL

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems (OS), Database Management Systems (DBMS)

Libraries & Frameworks: NumPy, Pandas, Matplotlib

Developer Tools: Git, GitHub, Google Colab, SQLite

Coding Profiles: Solved **400**+ DSA problems across LeetCode, Codeforces, and NeetCode

PROJECTS

In-Memory LRU Cache Engine

C++, Data Structures, Memory Management

- **Engineered a custom Least Recently Used (LRU) cache** from scratch in C++, seamlessly integrating a hash map with a doubly linked list to guarantee **O(1)** time complexity for all lookups, insertions, and evictions.
- **Optimized memory pointer manipulation** by implementing dummy sentinel nodes, systematically eliminating empty-list and single-node edge cases from the core eviction logic.
- **Developed an interactive CLI** featuring real-time hit/miss rate tracking and strict input validation to empirically verify the accuracy of the caching algorithm under dynamic capacity constraints.

Lossless Huffman Compression Engine

C++, Data Structures, Greedy Algorithms, Memory Management

- **Engineered a lossless text compression engine** in C++ implementing the Huffman coding algorithm, utilizing a custom Min-Heap and Hash Maps to dynamically assign prefix-free binary codes based on character frequencies.
- **Architected a custom binary tree structure** for encoding and recursive decoding, optimizing memory management via custom destructors to systematically prevent memory leaks.
- **Developed an interactive testing CLI** capable of processing edge cases (e.g., single-character strings) and dynamically computing string compression ratios in real-time.

Mini Blockchain Console Application

Python, SQLite, SQL, SHA-256, Cryptography

- **Architected a blockchain simulator** utilizing SHA-256 hashing and adjustable-difficulty Proof-of-Work (PoW) to establish an immutable, cryptographically secure ledger.
- **Engineered a pending-transaction pool (mempool)** to bundle entries into blocks during mining, and built a validation protocol to detect tampering or broken hash links.
- Migrated ledger persistence from flat-file JSON to a **SQLite backend**, designing a normalized schema to enable direct SQL querying over chain state.

POSITIONS OF RESPONSIBILITY

Elan & Envision, Techno-Cultural Fest, IIT Hyderabad

- **Sponsorship Coordinator:** Spearheaded corporate outreach and pipeline management, securing direct financial sponsorships and acting as a dedicated POC for assigned companies to ensure on-ground fulfillment of deliverables.

Finance and Consulting Club, IIT Hyderabad

- **Operations Deputy Coordinator:** Managed the end-to-end execution of pre-fest workshops and coordinated all logistical arrangements to ensure smooth operations during the main event.

E-Cell, IIT Hyderabad

- **Marketing Deputy Coordinator:** Spearheaded outreach initiatives to establish strategic collaborations and partnerships with Entrepreneurship Cells of various other institutes.

RELEVANT COURSES

Programming and Machine Learning Courses - Introduction to Programming in C language, Introduction to AI/ML

Mathematics - Linear Algebra, Applied Maths, Calculus, Numerical Methods, Differential Equation